# 7th Grade Math Exam — Detailed Solutions

All numerical results in this document were verified symbolically with SymPy via the bundled *math* skill.

## Part A — Proportionality & Proportion Coefficient

**Q1.**

a) Compute every ratio distance / time: 70/2 = 35, 105/3 = 35, 175/5 = 35, 245/7 = 35. All equal → proportional.

b) Proportion coefficient **k = 35** (km/h).

c) Formula: **d = 35 · t**.

**Q2.** With k = 3, so y = 3x.

x = 4 → y = 12 | y = 21 → x = 7 | x = 10 → y = 30.

| x | 2 | 4 | 7 | 10 |
|---|---|---|---|---|
| y | 6 | 12 | 21 | 30 |

**Q3.** Cross-multiplying 4/10 = x/25:

10 · x = 4 · 25 = 100 ⇒ **x = 10**.

**Q4.** Coefficient = 150 / 4 = 37.5 g per cookie.

For 18 cookies: 18 · 37.5 = **675 g**. (Cross-check: 18 · 150 / 4 = 2700 / 4 = 675.)

**Q5.**

Table a) ratios 3/1 = 6/2 = 9/3 = 12/4 = 3 → **proportional**, k = 3.

Table b) ratios 3, 2, 5/3 ≈ 1.67, 1.5 → **not proportional**. (In fact y = x + 2, an additive relation.)

## Part B — Percentages

**Q6.** 25% of 80 = (25/100) · 80 = **20**.

**Q7.** 15 / 60 = 1/4 = 25/100 = **25%**.

**Q8.** Multiplier 1 + 12/100 = 1.12. New price = 200 · 1.12 = **224 €**.

**Q9.** Multiplier 1 – 20/100 = 0.80. Final price = 50 · 0.80 = **40 €**.

**Q10.** 30% of 250 = (30/100) · 250 = (3/10) · 250 = **75 students**.

## Part C — Polynomials

Given $P(x) = 3x^2 + 2x - 1$, $Q(x) = x^2 - 5x + 4$.

**Q11.** Degree = **2**; leading coefficient = **3**.

**Q12.** Group like terms: $(3x^2 + x^2) + (2x - 5x) + (-1 + 4) = \mathbf{4x^2 - 3x + 3}$.

**Q13.** Distribute the minus: $3x^2 + 2x - 1 - x^2 + 5x - 4 = (3x^2 - x^2) + (2x + 5x) + (-1 - 4) = \mathbf{2x^2 + 7x - 5}$.

**Q14.** $2x \cdot (x + 3) = 2x \cdot x + 2x \cdot 3 = \mathbf{2x^2 + 6x}$.

**Q15.** $(x + 4)(x - 2) = x^2 - 2x + 4x - 8 = \mathbf{x^2 + 2x - 8}$.

**Q16.** $P(2) = 2^2 - 3\cdot2 + 5 = 4 - 6 + 5 = \mathbf{3}$.

## Bonus

Step 1: 80 · (1 – 25/100) = 80 · 0.75 = **60 €**. Step 2: 60 · (1 + 25/100) = 60 · 1.25 = **75 €**.

Final price = 75 €, **not** 80 €. The combined factor is 0.75 × 1.25 = 0.9375 < 1 — a net 6.25% loss. Equal percentage changes in opposite directions do not cancel because they are applied to different bases.